

INDIA SPACE ACADEMY
Summer Training Program 2026
Lab Report for Project (Part-1)

Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 Data

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Abstract

This report is a description of an automated pipeline for detecting and validating solar active regions using Level 1 narrowband image from the Solar Ultraviolet Imaging Telescope (SUIT) onboard India's Aditya-L1 spacecraft. The dataset used is a single FITS file acquired on May 9, 2026, in the Mg II k spectral line at 279.6 nm. The pipeline comprises flat-field correction, intensity normalization, disk masking, Otsu-based threshold segmentation, morphological artifact removal, cross-validation from external sources (sourced from Heliophysics Event Knowledgebase) along with histogram analysis of both raw and preprocessed image arrays for profound understanding of raw and preprocessed data. The pipeline was successful in identifying 7 unique NOAA active regions with 61 detected blobs thus demonstrating the ability of this pipeline to detect space weather phenomena. At the end limitations of this pipeline are also discussed alongside directions that can be used for future development. The limitations if resolved can improve the accuracy of the pipeline in detecting the active regions greatly.

OBJECTIVE: This project is aimed to build a pipeline to detect and validate solar active regions using Aditya L1's imagery of sun taken by the onboard SUT payload using 'NB03' filter. It covers preprocessing, thresholding, morphological operations and cross-validation with the NOAA active region catalog.

Dataset Information:

- File name:
SUT_T26_0649_002132_Lev1.0_2026-05-09T07.09.21.400_08B2NB03.fits
- Instrument: SUT, Narrowband filter 03 (NB03), Mg II k line (279.6 nm)
- Date and Time of observation: 9 May 2026 07:09:21 AM

The following date and time have been chosen because of the unavailability of the dataset that was mentioned in the pdf (31 Dec 2023).

METHODOLOGY

1. **Raw Image Visualization:** Before I started to work on the image I uploaded the raw FITS file and visualized it using SunPy in its unprocessed state. There were visible intensity variations in the disk and also the image was an off-centered image of the sun.

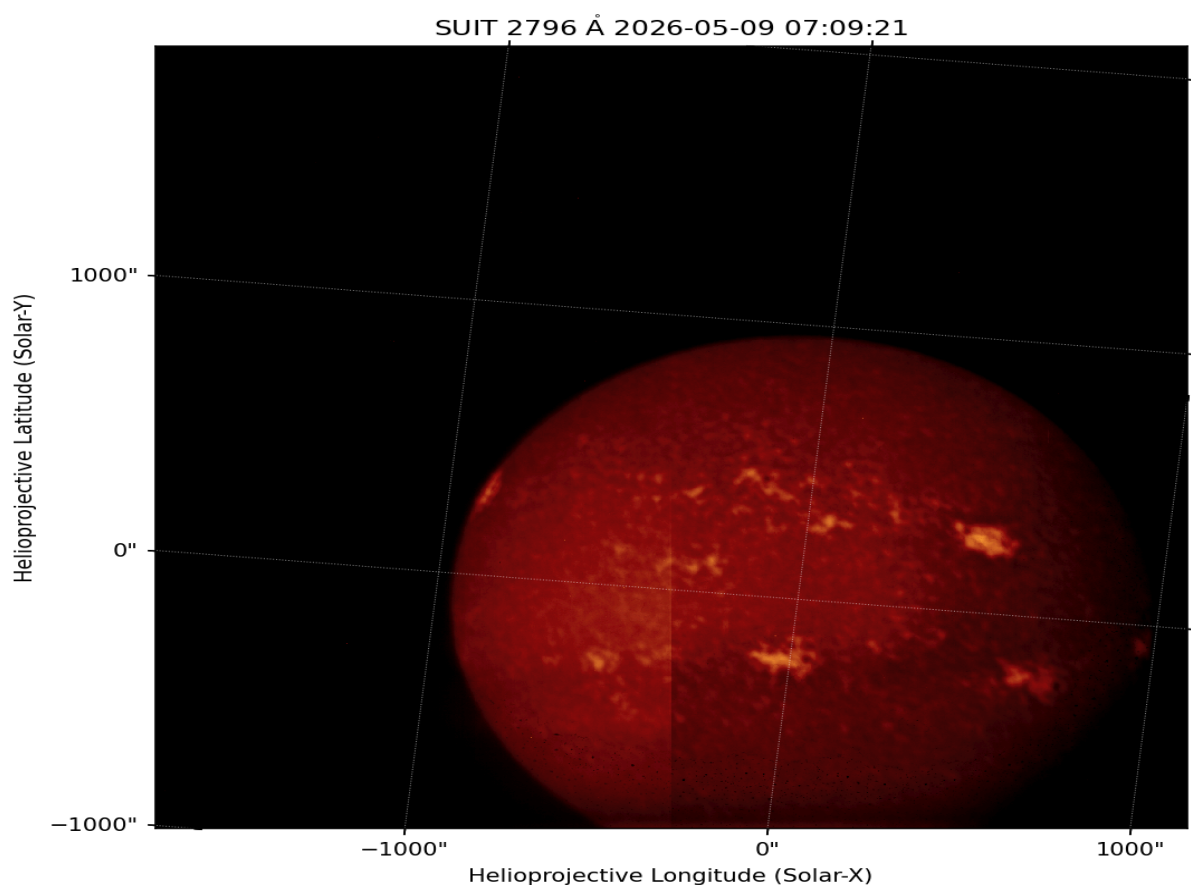


Figure 1-Raw Image

2. **Histogram Analysis:** I plotted the histogram of the raw image array to understand the data better. At the first look it very much looked like a bimodal distribution except for the peak at origin. This visualization gave us a valid reason to proceed with Otsu thresholding. The peaks correspond to physical significance which I have explained in detail in my notebook.

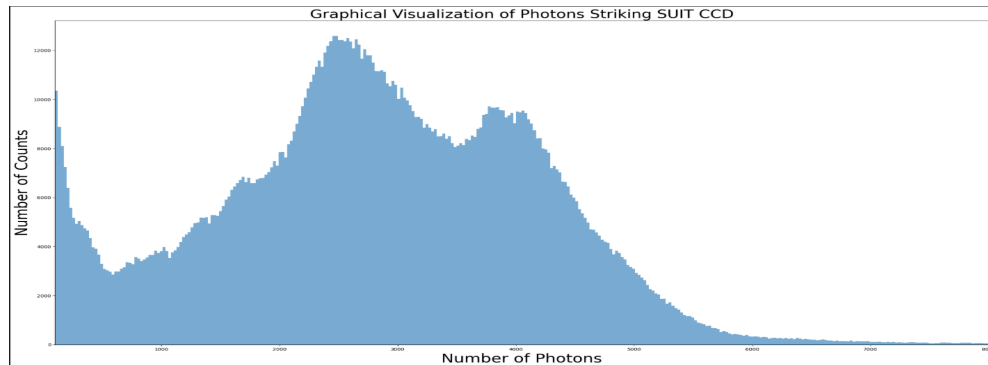


Figure 2-Raw Histogram

3. **Flat-field correction:** It was done to account for pixel-to-pixel sensitivity variations in the SUIT's detectors. It also smoothened the sudden rises and falls in the plot. An image of this then was generated which showed us very high off-disk distortions which were taken care of in the next steps.
4. **Normalization:** This step is essential for our analysis as it rescales the data into a standard range (in our case- 0 to 1). Percentile-Normalization was specifically chosen as it takes care of outliers by making them comparable to other smaller data points.
5. **Circular Masking:** A circular disk mask was then applied using parameters radius=775, x_center=1300, and y_center=650. It isolated the solar disk from the off-disk background. This reduced the off-disk artifacts substantially.

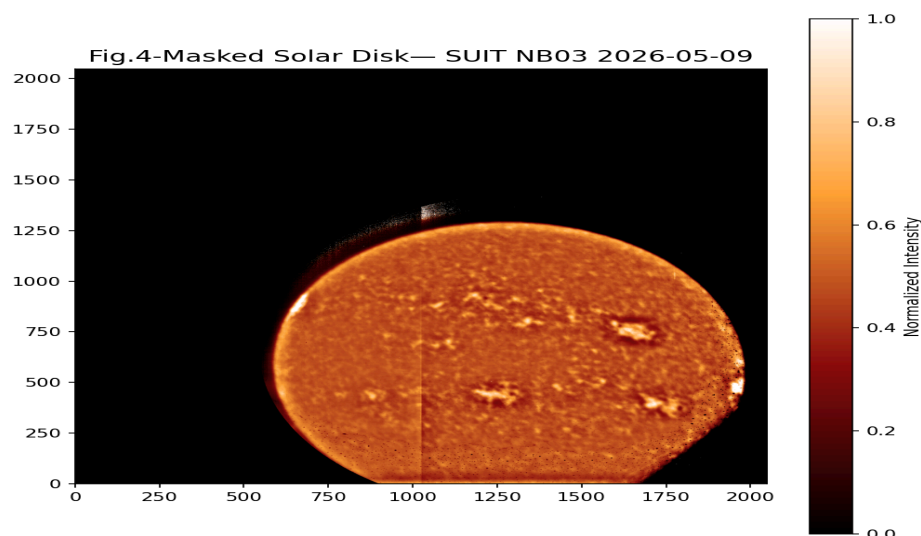


Figure 3-Masked Image

6. **Histogram Analysis:** A second histogram of masked and preprocessed array was then plotted to understand the changes brought about by our former steps. The histogram which was earlier a bimodal distribution, after preprocessing and masking lost its bimodality. Due to this loss of bimodality, the scope of error in Otsu's algorithm increased.

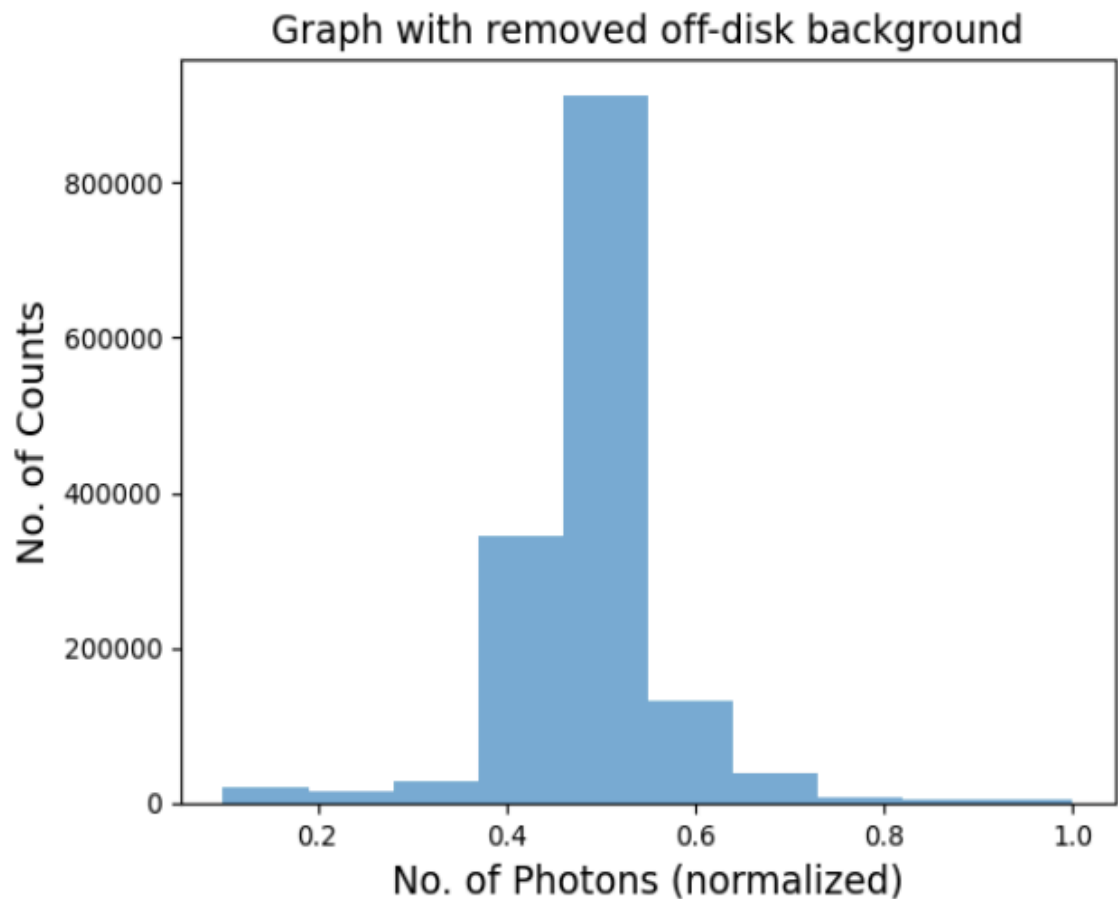


Figure 4-Preprocessed Histogram

7. **Otsu thresholding:** It was applied exclusively on disk pixels with intensity above 0.1 normalized value in order to prevent background leakage. A threshold value of approximately 0.5341 was calculated by Otsu. The values above this were classified as active regions and those lower than this as quiet sun.
8. **Morphological Operations:** Otsu threshold does only intensity based classification which led to identification of solar limb and off-disk distortions as active regions too. I tried to solve this issue by employing morphological

operations to use. Solar Limb, on disk horizontal patch, on-disk vertical patch and partially the off-disk distortion were removed using them.

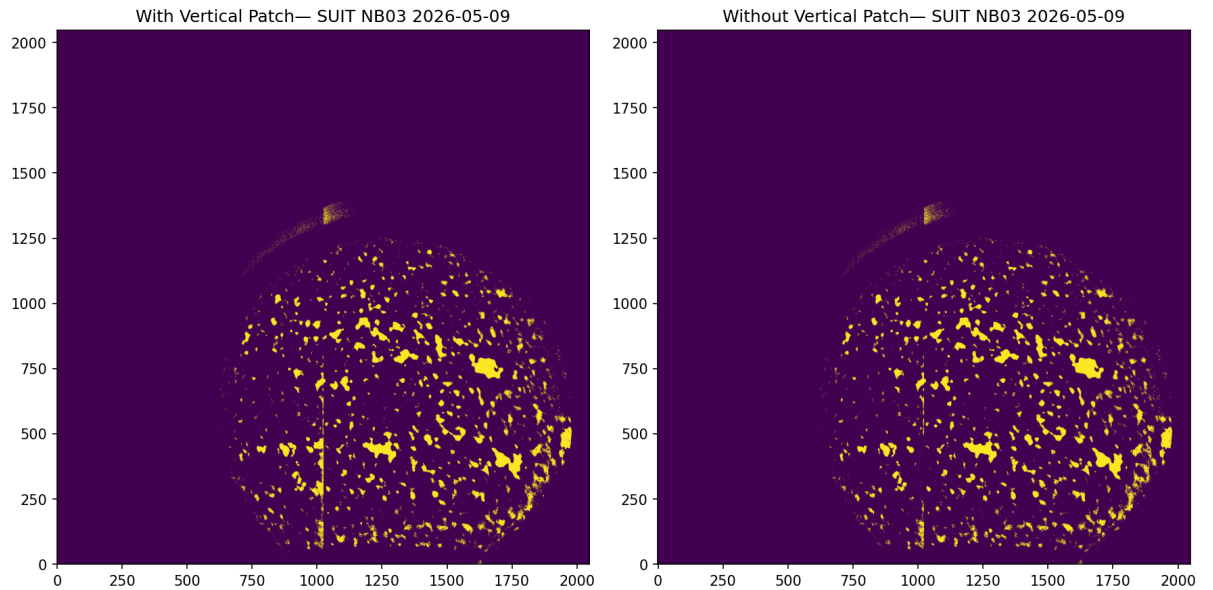


Figure 5-Final Image (Right) after morphological operations

9. **Final Visualization:** A final image was reached with contour overlay and false color map. A comparison is drawn between the final image and the image obtained after morphological operations. It is because the off-disk distortion appears unusually large due to the inbuilt property of `plt.contour()` to interpolate between grid points automatically.

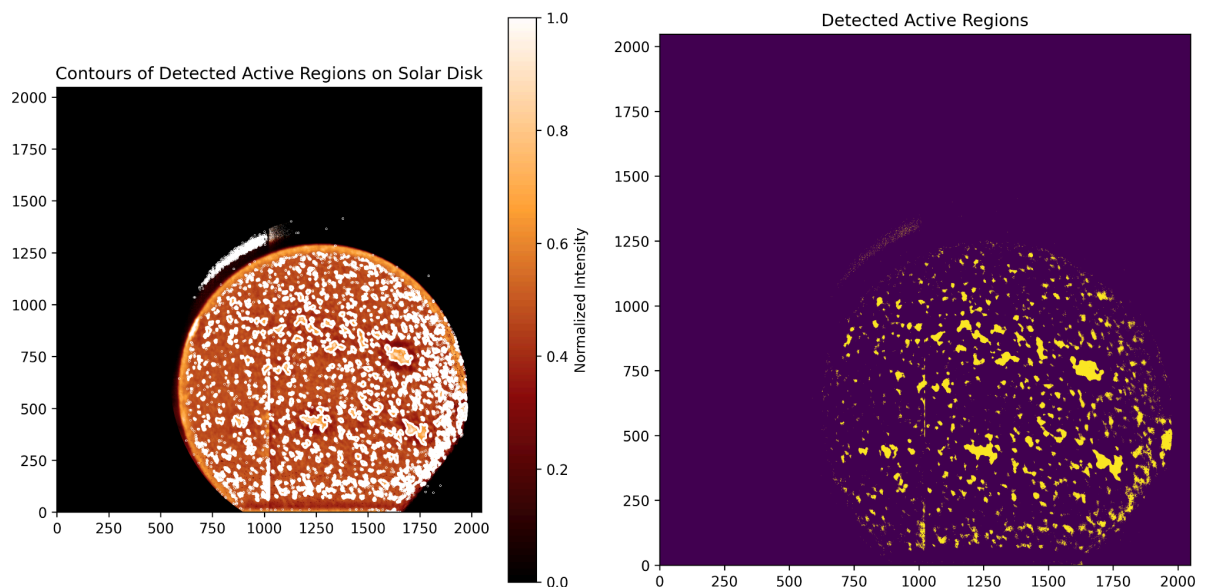


Figure 6-Final Visualization

10. **Cross Validation:** We cross-validated our findings with the NOAA catalog (extracted from HEK) of that same date. 7 unique NOAA ARs were confirmed with 61 blobs being verified. The reason for many blobs being assigned the same AR number was because of fragmentation of active regions into smaller active regions due to Otsu thresholding hence a single region was identified multiple times. The main reason for this was due to the loss of bimodality in our preprocessed array.

```
Unique NOAA ARs matched: {np.int32(14432), np.int32(14433), np.int32(14434), np.int32(14435), np.int32(14436), np.int32(14429), np.int32(14431)}  
Total verified blobs: 61
```

11. Discussion of results:

- Our pipeline detected **AR14436**. The next day (10 May 2026) it produced a M5.7 flare and a CME travelling at ~650 km/s.
- **AR14432** showed elevated C-class flare activity around the observation date with evolving magnetic polarity.

RESULTS:

1. The raw image reveals clear intensity variations across the solar disk (Figure 1). The bright patches in the disk are visually distinguishable. The corresponding histogram of the image array (Figure 2) displayed an almost bimodal distribution with peaks corresponding to physical significant phenomena.
2. After applying flat-field correction, normalization, and disk masking, the preprocessed histogram (Figure 4) showed a notably cleaner distribution with a single peak thus destroying the bimodality.
3. Otsu thresholding applied on disk pixels **above 0.1** normalized intensity yielded a threshold value of approximately **0.5341** thereby separating active regions and quiet sun.
4. After morphological cleanup which included the removal of the solar limb, a vertical patch, a horizontal patch and partial off-disk distortion— a total of 61 regions were identified across the solar disk (Figure 5).
5. On cross-validating, **7 unique NOAA active regions** were confirmed to be present on May 9, 2026. All **61** detected blobs were spatially assigned to these cataloged regions (Figure 6).
6. **NOAA AR14436**, detected in the pipeline output, was subsequently associated with an M5.7 class solar flare and a coronal mass ejection on May 10, 2026 — demonstrating the pipeline's capability to identify pre-eruptive active regions. **NOAA AR14432** was also identified in the pipeline which showed an elevated C-class flare activity with evolving magnetic polarity.

DISCUSSION:

The pipeline successfully detected and validated solar active regions in the 'SUIT NB03' image from May 9, 2026. However, several limitations were encountered throughout the process that warrant detailed discussion:

1. Flat-field correction, while necessary for detector calibration, is found to suppress active region contrast. Applying Otsu on the FFC-corrected data produced results which are segmented, and due to this, we are getting 61 detected blobs corresponding to only 7 regions.
2. The most significant source of limitation was Otsu thresholding. It has various limitations:
 - First of all, it is a method which is best suited for bimodal distribution. If the distribution doesn't follow bimodality (either having a single peak or more than 2 peaks) then it will detect a wrong threshold and hence cause significant error in classification. This happened when it was run directly on the fully masked array (including background pixels), it was found that the threshold shifted towards lower intensity. This was resolved by restricting the thresholding to disk pixels having normalized values greater than 0.1 which reduced background leakage. But this correction had a measurable impact on the final threshold value and eventually on detection.

A manual or somewhat more adaptive thresholding approach would have been a better choice.

 - Otsu thresholding considers segmentation on the basis of intensity based values only. Due to this, the solar limb too was detected as an active region. The solar limb was excluded from the solar disk by applying morphological operations which are discussed below in more detail.
3. Masking is another tool to which we resorted in clearing the off-disk distortions in the image of the sun. It worked very effectively as could be seen from the differences between unmasked and masked image but the process of approximating the center coordinates of disk and radius of disk was manual due to which a lot of time was consumed to arrive at the final result.
4. Morphological operations helped us in removing off-disk artifacts and on-disk patches without which our analysis would have been highly inaccurate. The data that I chose to operate on turned out to be an off-centered image of the sun thus being highly distorted in regions around western and southern

hemisphere (as warned in the PRADAN portal). These operations along with masking helped us minimize those distortions and on-screen patches. However, the process relied on manually verifying the coordinates again and again before reaching a suitable result hence costing a lot of time.

5. Despite all these limitations, a scientifically valuable finding emerged from the results of our pipeline. **NOAA AR14436** was detected by it on May 9, 2026 and it produced an M5.7 class solar flare and an associated coronal mass ejection on May 10, 2026. **AR14432** was also identified in the pipeline which showed an elevated C-class flare activity with evolving magnetic polarity.

Conclusion:

This project successfully developed an automated pipeline for detecting and validating solar active regions using the image data from Aditya L1's onboard SUIT (NB03 filtered image was considered for analysis). Starting from raw FITS data, the pipeline covered- raw histogram analysis, flat-field correction, normalization, disk masking, preprocessed histogram analysis, Otsu-based thresholding, morphological artifact removal, and cross-validation against the NOAA catalog which was sourced from HEK. A total of 61 blobs were detected, corresponding to 7 confirmed NOAA active regions on May 9, 2026. The detection of NOAA AR14436 — which subsequently produced an M5.7 flare and CME on May 10 and of AR14432 which showed an elevated C-class flare activity with evolving magnetic polarity — tells us the practical relevance of this pipeline. Limitations such as fragmentation due to Otsu thresholding and manual searching of coordinates for masking and morphological operations exist.

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